AI STACK COMPARISON

AWS Vs AZURE Vs GCP

AWS: AI Stack Overview

AWS provides a full-stack AI ecosystem—from raw data to real-time inference—designed for enterprise-scale workloads.

AWS AI Stack Breakdown

1. Data Foundation

- S3 (Storage)
- Glue (ETL)
- Redshift (Data Warehouse)

2. AI/ML Development

- SageMaker (Complete ML lifecycle)
- Deep Learning AMIs
- Bedrock (Foundation Model access)

3. Generative AI Tools

- Amazon Bedrock (Claude, Llama, Titan models)
- Bedrock Agents
- PartyRock (No-code app builder)

4. Deployment & MLOps

- SageMaker Pipelines, Feature Store, Model Registry
- Lambda / ECS / EKS for model serving

Why AWS Stands Out

Best ecosystem for enterprises building full ML pipelines with tight integration and massive scalability.

Azure:
AI Stack Overview

Azure is tightly integrated with Microsoft 365 and specializes in enterprise-grade AI, governance, and responsible AI tooling.

AWS AI Stack Breakdown

1. Data Foundation

- Azure Data Lake
- Synapse Analytics
- Azure Databricks

2. AI/ML Development

- Azure Machine Learning Studio
- Prompt Flow
- Responsible AI Dashboard

3. Generative AI Tools

- Azure OpenAI (GPT-4, GPT-4o, GPT-4.1, DALL·E)
- Copilot Studio
- Azure AI Foundry

4. Deployment & MLOps

- Managed Online Endpoints
- ML Ops with GitHub Actions
- AKS (Kubernetes)

Why Azure Stands Out

Best for companies already using Microsoft products and wanting seamless integration with Copilot, Office, and enterprise governance.

GCP:
AI Stack Overview

GCP is built around data analytics, scalable training, and cutting-edge generative AI through Vertex AI.

GCP AI Stack Breakdown

1. Data Foundation

- BigQuery
- Cloud Storage
- Dataform / Dataproc

2. AI/ML Development

- Vertex AI (unified ML platform)
- AutoML
- Notebook Instances

3. Generative AI Tools

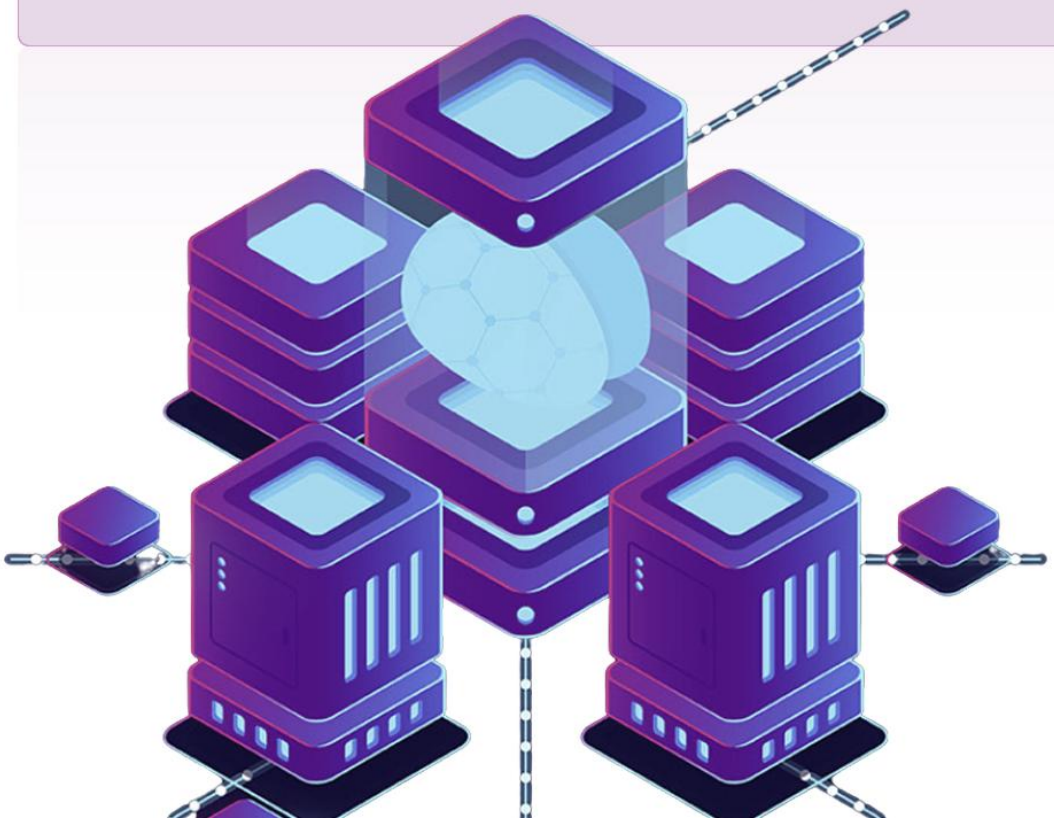
- Gemini Models
- AI Studio
- Model Garden (3rd-party models)

4. Deployment & MLOps

- Vertex AI Pipelines
- Endpoints for deployment
- GKE for scalable serving

Why GCP Stands Out

Best for data-heavy AI workloads, analytics-driven teams, and orgs building with Gemini or advanced MLOps tools.



Final Comparison (Simple)

Layer	AWS	Azure	GCP
Data	S3 + Redshift	Data Lake + Synapse	BigQuery
ML Platform	SageMaker	Azure ML	Vertex AI
Gen AI	Bedrock	Azure OpenAI	Gemini
Best For	Enterprise ML at scale	Microsoft ecosystem	Data + GenAI workloads

AI stacks keep evolving fast. Build skills across at least one cloud provider and stay adaptable for multi-cloud AI environments.